M5h: magnetohydrostatic (MHS) reconstruction of the full Bifrost cube — what is solved and how
HSEMHS (Julia), run `out/m5h/run_fullz`, 22 Oct 2026; grid $504 \times 504 \times 496$ (uncropped), EOS table `cache/eos_table_fullz.h5`

1. Physical problem

Given the Bifrost temperature $T(x,y,z)$ (held fixed) and the Bifrost field $\mathbf{B}_{\rm obs}$ with uncertainty σ, find the gas pressure P and a field $\mathbf{B}$ that satisfy magnetostatic force balance

$$-\nabla P + \mathbf{J}\times\mathbf{B} + \rho\,\mathbf{g} = 0, \qquad \mathbf{J} = \nabla\times\mathbf{B}/\mu_0, \qquad \nabla\cdot\mathbf{B} = 0, \qquad \rho = \rho_{\rm EOS}(T,P), \qquad \mathbf{g} = -274\,\hat{\mathbf{z}}\ \mathrm{m\,s^{-2}},$$

while staying as close to $\mathbf{B}_{\rm obs}$ as σ allows. ρ (and the opacity χ_{500}) come from the Wittmann EOS table (T 2172 K–1.781 MK, $\log_{10} P$ $-13\ldots15$). Periodic in x, y; $\sigma = 5\%\,|\mathbf{B}_{\rm obs}| \oplus 5\,\mathrm{G}$.

2. Unknowns and discretisation (staggered, finite volume)

State $\mathbf{x} = (u, \tilde{\mathbf{A}}, \bar{b})$: $u = \ln P$ at cell centres; vector potential $\mathbf{A} = A_{\rm ref}\tilde{\mathbf{A}}$ on cell edges; one scalar $\bar{b}$ (mean B_z). $\mathbf{B} = \nabla\times\mathbf{A} + \bar{b}\,\hat{\mathbf{z}}$ on faces, so $\nabla\cdot\mathbf{B} = 0$ holds exactly. Pressure gauge: the area-weighted mean of $\ln P$ on the bottom plane is fixed at $\ln P_{\rm ref}$, $P_{\rm ref} = 5.222\times10^6$ Pa. About 5×10^8 unknowns.

3. Objective (least squares)

$$\Phi = \underbrace{\frac{\lambda_F}{N_F}\sum_{\rm faces} w_f \frac{R_f^2}{S_f^2}}_{\Phi_F\ \text{force balance}} + \underbrace{\frac{\lambda_F\lambda_Q}{N_F}\sum_{\rm edges} w_e\Big(\frac{B_{\rm ref}\,\frac{1}{4}\delta^2 J_e}{S_e}\Big)^2}_{\Phi_Q\ \text{grid-scale currents}} + \underbrace{\frac{1}{N_B}\sum_{\rm faces}\frac{(B_f - B_{{\rm obs},f})^2}{\sigma_f^2}}_{\Phi_B\ \text{data}} + \underbrace{\frac{\lambda_S}{N}\sum w\sum_a \frac{(h_a^2\partial_a^2\,\delta\mathbf{B})^2}{\sigma^2}}_{\Phi_S\ \text{smooth misfit}}$$

R_f = face component of $\nabla P + \rho g\hat{\mathbf{z}} - \mathbf{J}\times\mathbf{B}$ (horizontal faces of layers $2\ldots N_z-1$, interior z-faces); $S = \max(\rho_0 g,\ 0.01\,\mathrm{median}_k\,\rho_0 g)$ from the initial state (fixed); w = cell-volume weights. Weights used: $\lambda_F = 10^{-4}$, $\lambda_Q = 10^{-3}$, $\lambda_S = 10^{-3}$, $\lambda_U = 0$. The log prints the terms as (F, $N{+}Q$, B, S, U).

4. How it is solved

1. **Initial pressure:** hydrostatic integration of each column with the fixed T and the EOS, anchored to $P_{\rm ref}$.
2. **Initial field** (`init_A!`): pressure frozen, $\lambda_F = \lambda_Q = 0$, so $\Phi_B + \Phi_S$ is quadratic in $(\tilde{\mathbf{A}}, \bar{b})$; solved by CG on the normal equations (capped at 3000 iterations; reached residual ratio 1.2×10^{-6}).
3. **Joint trust-region Gauss–Newton** on all of $(u, \tilde{\mathbf{A}}, \bar{b})$. Each outer iteration:
   - solve $(H_{\rm GN} + \mu D)\,\delta = -\nabla\Phi$ by Steihaug–Toint preconditioned CG (at most 100 iterations, relative tolerance 10^{-2}) inside a D-norm trust ball. D is the preconditioner diagonal, with the pressure block weighted $\times100$.
   - pressure step projected onto the bounds: $|\Delta\ln P| \le 5$ per node per step ($\Delta P/P \le e^5 - 1 \approx 147$), $\ln P \ge \ln P_{\rm init} - 3$, bottom anchor unchanged; projected-Cauchy safeguard.
   - accept if $\rho = \dfrac{\text{actual decrease}}{\text{predicted decrease}} > 10^{-4}$. If $\rho > 0.75$ the radius doubles and μ is divided by 3; if $\rho < 0.25$ the radius shrinks and μ is multiplied by 4.
4. **Parallel:** MPI, 24 ranks (6×4 in x, y; z not split) $\times$ 16 threads on hyades4/5/15/21; checkpoints are written by cooperative HDF5 (the ranks write their blocks in turn).

5. What is measured — convergence criterion (M5h)

After every accepted iteration n, for every column (i,j):

$$\tau_{500}(z) = \tau_{\rm top} + \int_z^{z_{\rm top}} \chi_{500}\big(T, P^{(n)}\big)\,dz', \qquad \tau_{\rm top} = \chi_1 H_\chi\ (H_\chi < 5\,\mathrm{Mm}), \qquad z_{\tau1}^{(n)}(i,j):\ \log_{10}\tau_{500} = 0.$$

$$dz_{\max}^{(n)} = \max_{i,j}\big|z_{\tau1}^{(n)} - z_{\tau1}^{(n-1)}\big| \ <\ 1\ \mathrm{mm} \quad \text{on 3 consecutive accepted iterations, with all 254 016 columns having a valid crossing.}$$

Also logged every iteration (`history.csv`):

- Φ and each of its terms; raw force residual; RMS and 99th percentile of $|R|/S$;
- worst normalised data residual $|B - B_{\rm obs}|/\sigma$ on faces and at centres;
- $\max\Delta P/P$ and $\max|\Delta\mathbf{B}|$ of the step; trust ratio ρ;
- $z_{\tau1}$ min/median/max; counts of missing, non-finite and EOS-failed columns.

6. Status (17:48, accepted iteration 8)

- Φ: $1.366\times10^5 \to 1.374\times10^4$ (almost entirely Φ_F).
- Columns: all 254 016 valid, no EOS failures.
- $dz_{\max} = 50$ km; median $z_{\tau1}$ is moving by ±30–40 km between iterations. Still far from 1 mm.
- Cost: 4–6 min per accepted iteration; each checkpoint save takes about 18 min.